

SAVR: Scooping Adaptation for Variable food properties via Reinforcement Learning

J-Anne Yow^{1,2}, Wei Tech Ang^{1,2}

¹School of Mechanical and Aerospace Engineering, Nanyang Technological University Singapore

²Singapore-ETH Centre, Future Health Technologies Programme

Abstract—Personalizing bite sizes is crucial for robot-assisted feeding, as users have diverse dietary needs and preferences. However, precisely controlling the amount of food scooped remains a challenge due to variations in food properties, such as texture, granularity and cohesion. This work introduces SAVR (Scooping Adaptation for Variable food properties via Reinforcement learning), a learning-based framework that enables robots to scoop a targeted amount of food while adapting to different food characteristics. SAVR integrates Dynamic Motion Primitives (DMPs), with Reinforcement Learning (RL), where DMPs provide a structured motion representation, and RL refines execution by modifying the force term within the DMP formulation. This formulation enables efficient learning by allowing the RL agent to fine-tune the scooping trajectory rather than learning entire trajectories from scratch. Through ablation studies, we show that segmented spoon and food masks, combined with force-torque data, are essential for accurate scooping, significantly improving sim-to-real transfer. We validate SAVR on a real robotic system, demonstrating substantial improvements in accuracy and adaptability over baselines, without any additional fine-tuning.

Index Terms—robot-assisted feeding, personalized feeding, reinforcement learning, assistive robotics,

I. INTRODUCTION

In assistive feeding, robots must adapt to various user preferences, including bite sequence, feeding speed, and bite transfer approach, to create a comfortable and intuitive dining experience. Among these factors, bite-size personalization is particularly important [1], as users have different portion size preferences based on their chewing ability, swallowing capacity, and dietary requirements. For instance, individuals with dysphagia may require smaller, controlled portions to prevent choking, while others may prefer larger bites for a more natural or faster dining experience. A robot capable of dynamically adjusting the scooped amount based on food properties and user preferences would significantly enhance the practicality and usability of robotic feeding systems.

While prior works have explored personalization in feeding strategies [2]–[5], precise control over the amount of food scooped remains largely unaddressed. Previous work has explored bite-size personalization, but primarily focused on mapping user preferences expressed in language to motion

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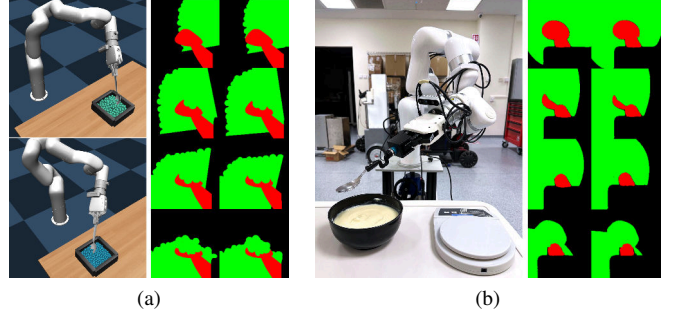


Fig. 1. Overview of the simulation and real-world environments, along with the image observations used for learning. (a) Simulation environment with randomized particle properties and food poses for domain randomization. (b) Real-world setup. Segmented spoon and food masks help reduce the sim-to-real gap by providing consistent visual cues across domains. Only a subset of images from the feeding sequence is shown.

primitive parameters [6]. The reliance on standard Dynamic Motion Primitives (DMPs) to adjust bite-size struggles with real-world variability in food properties due to its open-loop nature, leading to inconsistent bite sizes, as demonstrated in our experiments. In contrast, human caregivers instinctively adjust their scooping movements based on the food’s texture and the desired portion size, an ability that robotic feeding systems must emulate for true personalization.

Existing robotic scooping works have explored handling different food textures [7]–[9], but these approaches prioritize successful food acquisition rather than precisely controlling the scooped amount. Without this capability, inconsistencies in bite sizes may arise, limiting a robot’s ability to personalize feeding effectively. While prior work has attempted targeted scooping of water [10], it does not address the challenges of scooping solid and deformable foods or adapting to varying food properties. The ability to dynamically adjust scooping actions based on both food properties and target amount remains an open challenge, highlighting the need for an approach that enables both adaptability and precise amount control.

To address this, we introduce SAVR (Scooping Adaptation for Variable food properties via Reinforcement learning), a learning-based framework that enables goal-conditioned food scooping, allowing users to specify a target amount while adapting to different food properties. Inspired by human

behaviour, SAVR builds upon Dynamic Motion Primitives (DMPs) to provide a structured motion policy while leveraging Reinforcement Learning (RL) to fine-tune the scooping trajectory to achieve precise food acquisition. A key insight of our work is that leveraging segmented spoon and food masks with force-torque data significantly improves sim-to-real transfer. By focusing on spoon area segmentation, the model estimates the scooped amount in a way that is robust to domain gaps between simulated and real food textures, reducing reliance on variable visual cues. Combined with force sensing, this enables SAVR to generalize across diverse food properties, ensuring more accurate and reliable scooping.

We evaluate our framework both in simulation and real-world experiments, comparing it against baselines such as DMP-based scooping and RL without DMP priors. Our results demonstrate that SAVR significantly outperforms these baselines, achieving higher accuracy, improved adaptability, and better sim-to-real transfer in scooping tasks.

The key contributions of this work are:

- 1) We introduce the problem of goal-conditioned food scooping, enabling a robot to adjust scooped amounts based on target portion sizes and varying food properties.
- 2) We propose SAVR, integrating Dynamic Motion Primitives (DMPs) with Reinforcement Learning (RL) to adapt scooping actions based on food properties and user-specified targets. Real-world validation demonstrates SAVR's strong sim-to-real transfer, outperforming baselines in accuracy and adaptability across diverse food types.
- 3) We show that leveraging segmented spoon and food masks with force sensing is critical for closing the sim-to-real gap, enabling robust generalization to real-world food textures without the need for any additional fine-tuning.

II. RELATED WORK

A. Robotic Food Scooping

Several works have explored robotic food acquisition using a spoon, ladle, or fork for scooping [7]–[20]. While significant progress in scooping has been made, most approaches often do not consider adaptability to different food types, container configurations, and scooping a targeted amount of food.

Recent works have aimed to improve generalization in scooping by incorporating active perception. Tai *et al.* [7] introduced SCONE, which contains a pre-interaction sequence using active perception to predict food properties, enhancing generalization to different food types. However, their approach struggled with variations in bowl sizes, materials and positions, limiting its robustness. Liu *et al.* [8] later proposed a visual imitation network capable of handling diverse food types and bowl configurations. Ha *et al.* [20] presented a real2sim2real approach for acquiring soft food items using a fork. However, these approaches do not consider the amount of food scooped, treating the task as successful as long as food is acquired without spillage.

Some works have addressed controlling the scooped amount but remain limited in scope. Niu *et al.* [10] presented GOATS, an RL-based algorithm leveraging curriculum learning and reward shaping to scoop a targeted amount of water. However, their approach focused only on water scooping, and had some challenges bridging the sim-to-real gap, especially when dealing with water tanks of different sizes. Kageyama *et al.* [19] presented a predictive model to estimate the amount of food scooped using tactile sensors and adjust the motion accordingly to scoop the target weight. While their approach addressed quantity estimation, it did not account for different food properties.

Unlike prior methods that focus solely on food acquisition or estimating scooped weight, our work explicitly addresses adaptive scooping of a targeted food amount across diverse food properties.

B. Learning from Demonstration (LfD)

Learning from Demonstration (LfD) enables robots to learn tasks by imitating expert behavior. A widely used approach in LfD is Dynamic Motion Primitives (DMPs), which encode demonstrated trajectories in a non-linear system of differential equations [21]:

$$\tau \dot{z} = \alpha_z(\beta_z(g - y) - z) + f(\phi) + C_f \quad (1)$$

$$\tau \dot{y} = z \quad (2)$$

where τ is a time constant, α_z and β_z are positive constants that define the system's dynamics. The term $f(\phi)$ is a non-linear forcing function composed of a linear combination of N radial basis functions, enabling the robot to follow a smooth trajectory from the initial position y_0 to the final configuration g . ϕ is the phase variable governing the evolution of the trajectory over time, and C_f represents the external force term, which can be adjusted to modify the trajectory dynamically.

DMPs have been widely used in manipulation and adaptation tasks. Standard DMPs execute learned trajectories in an open-loop manner without real-time sensory feedback. To incorporate feedback, Pastor *et al.* [22] introduced the external force term C_f into DMPs to adapt movements online during robotic grasping. However, this approach assumes consistent force interactions, which does not hold when scooping food of varying properties [14].

Recent advances integrate learning-based approaches into LfD to improve adaptability. Hierarchical Neural Dynamic Policies (HNDP) [15] use RL to learn the radial basis function (RBF) weights for DMPs, enabling end-to-end policy learning from high-dimensional image inputs, but did not explicitly control the scooped amount or adapt to food properties. Residual reinforcement learning from demonstrations (RLFD) [23] applies task-space corrections to a base policy, showing success in contact-rich tasks like peg-in-hole insertion. However, in food scooping, where food properties vary significantly, such residual corrections may be insufficient for handling food-dependent variations in scooping.

Instead of using RL to adjust RBF weights for shaping trajectories or learning residual corrections, we directly modulate the force term C_f in the DMP formulation. This enables adaptive adjustments within the trajectory generation process itself, allowing the robot to dynamically refine its motion based on food properties.

III. APPROACH

We aim to develop a goal-conditioned policy that enables the robot to scoop a targeted amount of food while adapting to different food properties. To achieve this, we introduce Scooping Adaptation for Variable food properties via Reinforcement learning (SAVR), which combines Dynamic Motion Primitives (DMPs) with Reinforcement Learning (RL) for adaptive scooping.

DMPs handle variations in food poses by adjusting their start and goal states but cannot adapt to different food textures. To address this, we use RL to modify the DMP execution by applying corrective forces, allowing the system to dynamically adjust its motion based on observations.

The DMP runs at a higher frequency to manage precise motion, while the RL policy operates at a lower frequency, focusing on adjusting the force. This setup simplifies learning; without DMPs, training an RL policy directly in Cartesian space for targeted scooping would require extensive reward shaping and a significantly larger search space.

A. Problem Formulation

We define the problem of scooping a targeted amount of food as a finite-horizon discounted Markov decision process (MDP) $M = \langle S, A, T, R, \gamma \rangle$. At each timestep t , the agent observes the state $s_t \in S$, takes an action $a_t \in A$ based on policy $\pi_p(s_t)$ and receive a reward r_t based on the reward function $R(s_t, a_t)$. The objective is to find an optimal policy π_p^* that maximises the expected sum of discounted over T timesteps.

1) *State*: The state space S includes the following:

- History of end-effector poses, s^{eff} : A sequence of the most recent k end-effector poses, each represented as a 7-dimensional vector (position, quaternion) $s^{eff} \subseteq \mathbb{R}^{7 \times k}$. This history is included since the RL policy operates at a lower frequency than the DMP policy, so we provide more context for decision-making.
- History of force-torque sensor data, s^{ft} : A sequence of the most recent k force-torque measurements $s^{ft} \subseteq \mathbb{R}^{6 \times k}$.
- Cropped and masked image of the food and spoon, s^{img} : A $3 \times 200 \times 200$ tensor captured by a camera mounted on the robot's effector. The image is cropped to focus on the region containing the food and spoon, with separate identifiers (IDs) in the masks to distinguish between the spoon and food (Fig. 1).
- Target amount to be scooped, $s^{amt,*}$ and actual amount scooped, s^{amt} . The actual amount scooped s^{amt} is only updated at the final timestep of the episode.

- DMP step index, s^{dmp} : Indicates the current point in the scooping trajectory, which the policy uses to track its progress through the task.

2) *Action*: The action space A is continuous, ranging between $[-1, +1]$, representing a force applied in the z-direction to perturb the DMP policy. If no action is applied, i.e. $a = 0$ throughout the whole episode, an unperturbed DMP policy is executed. The RL agent's action a_t is applied directly to C_f in Equation 1, i.e. $C_f = a_t$. The agent can perform exploratory actions to achieve the target amount by modifying the trajectory through this external force.

3) *Reward*: A sparse reward function is used, providing feedback only at the final step of the episode based on the amount of food scooped. We define the target amount as a percentage of the maximum volume of particles that can be scooped, where a value of 100 represents the maximum possible amount. The reward function is shown in Equation 3, where δ represents the absolute difference in the target amount $s^{amt,*}$ and actual amount s^{amt} , i.e. $\delta = |s^{amt,*} - s^{amt}|$.

$$R(s_T, a_T) = \begin{cases} 10 & \text{if } \delta < 10 \\ -10 & \text{if } s^{amt} = 0 \\ 10/(1 + \delta/10) & \text{otherwise} \end{cases} \quad (3)$$

IV. SIMULATION EXPERIMENTS

Before deploying our approach in real-world experiments, we first evaluated SAVR in a controlled simulation environment. The primary goals of these simulation experiments were to assess the adaptability of SAVR in scooping varying amounts of food across different food properties, compare SAVR against baseline methods, and analyze the impact of different sensor modalities, through ablated versions of SAVR.

A. Simulation Setup

The robot policies were trained and tested in a simulated environment using MuJoCo [24] with the xArm-6 robotic arm (UFactory, China) (Fig. 1a). The simulation environment was designed to model realistic food scooping dynamics, incorporating variability in food properties and container configurations. At the start of each episode, the food and container poses, the volume of food in the container, and particle properties were randomized at the beginning of each new episode. To model different food properties, particles with varying sizes, masses, damping ratios, and friction loss were generated, simulating different textures and densities of real food. We also randomized the target amount to scoop, ranging from 10% to 100% of the maximum amount the spoon could physically scoop in a single action, in increments of 10%.

B. Implementation Details

The Dynamic Motion Primitive (DMP) was learned using a single scooping demonstration, which was sufficient to generate a successful baseline policy for acquiring food. However, the amount scooped fluctuated significantly due to variability in food properties, even when using the same DMP parameters.

TABLE I
PPO ALGORITHM HYPERPARAMETERS

Minibatch size	128	Learning rate	0.0003
Clipping parameter	0.2	Number of epochs	10
GAE lambda	0.95	Number of steps	256
Entropy coefficient	0.001	Target KL	0.01
Discount factor gamma	0.9	Optimizer	Adam

A position controller was used to execute the robot actions. The controller and DMP policy run at 100 Hz, while the RL policy runs at 20 Hz. To bridge the frequency gap and provide more context for decision-making, we include a history of $k = 5$ timesteps for both the end-effector pose and force-torque sensor data. Each episode consists of $T = 20$ timesteps, corresponding to the length of the DMP trajectory. For training, we used the Proximal Policy Optimization (PPO) [25] algorithm, implemented via Stable-Baselines3 [26]. The PPO hyperparameters used are shown in Table I.

Each algorithm was trained using three different random seeds. For evaluation, we tested each model over 20 episodes per target amount. The final results are reported as the average performance across all three seeds, with food properties randomized in the same manner across all methods to ensure consistency.

C. Evaluation Metrics

To evaluate the effectiveness of our approach, we define task success as scooping an amount of food within 10% of the target amount. Performance is assessed based on two key metrics: (1) Success rate: The percentage of trials where the robot successfully scooped within 10% of the target amount; (2) Amount error: The absolute difference between the actual scooped amount and the desired target amount.

D. Baselines in Simulation

To compare the effectiveness of **SAVR**, we compared it against the following approaches:

- **DMP**: A learning from demonstration approach that serves as the motion primitive foundation for our method.
- **PPO-NoPrior**: A state-of-the-art on-policy deep RL algorithm, where the agent directly controls end-effector pose adjustments without leveraging DMPs.
- **SAVR-ForceOnly**: An ablated version of our approach without any image observations.
- **SAVR-RGBImage+Force**: An ablated version of our approach using RGB images and force observations instead of segmented images.
- **SAVR-SegImageOnly**: An ablated version of our approach using only segmented images of the spoon and food, without force sensing.

All methods use the same hyperparameters in Table I for consistency.

E. Results in Simulation

Tables II and III summarize the success rates and amount errors for each method. Each algorithm was evaluated on

TABLE II
OVERALL RESULTS IN SIMULATION

Approach	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$
DMP	18.50 $\pm$ 0.19	31.31 $\pm$ 22.74
PPO	0.00 $\pm$ 0.00	55.0 $\pm$ 28.75
SAVR (Ours)	57.33 $\pm$ 0.08	10.77 $\pm$ 8.70
SAVR-RGBImage+Force	35.67 $\pm$ 0.08	18.19 $\pm$ 14.74
SAVR-ForceOnly	53.67 $\pm$ 0.08	11.79 $\pm$ 9.36
SAVR-SegImageOnly	51.17 $\pm$ 0.08	11.57 $\pm$ 8.71

Note: Bold indicates the highest success rate and lowest average error. SR denotes success rates, defined as scooping within 10% of the target amount; SD denotes standard deviation.

20 trials per target amount, with results averaged across three trained policies using different random seeds. The final reported values represent the mean performance across all seeds and trials.

The results indicate that our proposed framework and its ablated variants consistently outperform the baseline methods (DMP and PPO-NoPrior) across target levels. Notably, PPO-NoPrior failed to learn a valid scooping strategy, resulting in zero success across all target levels, with the highest average error of 55%. DMP struggled to reliably achieve the target amount, with relatively low success rates and high amount errors due to its inability to adjust to food property variations. SAVR achieved the overall highest success rates and lowest amount errors, demonstrating that the proposed approach significantly improved scooping precision. Standard deviation analysis reveals that SAVR is more consistent across trials, with a lower variance in the scooped amount compared to baselines. This suggests that SAVR’s adaptive approach leads to both higher accuracy and more stable performance.

Among the ablated versions of SAVR, SAVR-RGBImage+Force underperformed compared to SAVR-ForceOnly and SAVR-SegImageOnly. This suggests that incorporating RGB image information with force data may introduce noise rather than improving adaptation, while segmenting and masking the image helped the model learn better. Both SAVR-ForceOnly and SAVR-SegImageOnly achieved success rates and error metrics comparable to the full SAVR framework, suggesting that force sensing only or segmented image observations only also perform well in simulation.

While the overall success rates of SAVR (57%) may seem modest, we note that success is defined as scooping within 10% of the target amount, which is a strict evaluation criterion given the variability in food properties. The results demonstrate that adaptive control strategies significantly improved scooping accuracy compared to fixed motion primitives (DMP) and non-adaptive RL policies (PPO-NoPrior).

However, it remains unclear whether these trends hold in real-world experiments, particularly in physical conditions where force sensing and perception errors may affect performance differently. To address this, we validate SAVR and its key ablations on a real robotic system.

TABLE III
SIMULATION RESULTS FOR EACH TARGET AMOUNT (10%–100%)

Approach	10% Target		20% Target		30% Target		40% Target		50% Target	
	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$
DMP	0.00 $\pm$ 0.00	61.03 $\pm$ 22.42	0.00 $\pm$ 0.00	55.79 $\pm$ 23.27	10.00 $\pm$ 1.54	35.90 $\pm$ 17.21	5.00 $\pm$ 1.12	32.03 $\pm$ 19.00	10.00 $\pm$ 1.54	29.44 $\pm$ 18.54
PPO-NoPrior	0.00 $\pm$ 0.00	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	20.00 $\pm$ 0.00	0.00 $\pm$ 0.00	30.00 $\pm$ 0.00	0.00 $\pm$ 0.00	40.00 $\pm$ 0.00	0.00 $\pm$ 0.00	50.00 $\pm$ 0.00
SAVR (Ours)	76.67 $\pm$ 0.71	8.42 $\pm$ 7.52	70.00 $\pm$ 0.77	8.94 $\pm$ 7.66	66.67 $\pm$ 0.79	7.96 $\pm$ 7.39	63.33 $\pm$ 0.81	9.38 $\pm$ 6.09	55.00 $\pm$ 0.84	9.94 $\pm$ 6.80
SAVR-RGBImage+Force	56.67 $\pm$ 0.83	13.99 $\pm$ 14.94	55.00 $\pm$ 0.84	13.28 $\pm$ 12.89	45.00 $\pm$ 0.84	11.93 $\pm$ 7.56	40.00 $\pm$ 0.82	13.61 $\pm$ 9.98	36.67 $\pm$ 0.81	15.56 $\pm$ 9.71
SAVR-ForceOnly	90.00 $\pm$ 0.50	6.86 $\pm$ 6.56	60.00 $\pm$ 0.82	9.59 $\pm$ 6.88	41.67 $\pm$ 0.83	12.26 $\pm$ 7.29	56.67 $\pm$ 0.83	10.39 $\pm$ 6.56	56.67 $\pm$ 0.83	11.71 $\pm$ 9.03
SAVR-SegImageOnly	86.67 $\pm$ 0.57	6.30 $\pm$ 6.96	73.33 $\pm$ 0.74	7.31 $\pm$ 5.15	56.67 $\pm$ 0.83	8.87 $\pm$ 5.43	51.67 $\pm$ 0.84	10.09 $\pm$ 7.74	43.33 $\pm$ 0.83	13.57 $\pm$ 9.34

Approach	60% Target		70% Target		80% Target		90% Target		100% Target	
	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Avg Error $\pm$ SD (%) $\downarrow$
DMP	50.00 $\pm$ 2.56	15.42 $\pm$ 11.98	30.00 $\pm$ 2.35	20.45 $\pm$ 16.80	40.00 $\pm$ 2.51	16.05 $\pm$ 11.14	15.00 $\pm$ 1.83	22.04 $\pm$ 12.66	25.00 $\pm$ 2.22	24.95 $\pm$ 16.69
PPO-NoPrior	0.00 $\pm$ 0.00	60.00 $\pm$ 0.00	0.00 $\pm$ 0.00	70.00 $\pm$ 0.00	0.00 $\pm$ 0.00	80.00 $\pm$ 0.00	0.00 $\pm$ 0.00	90.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100.00 $\pm$ 0.00
SAVR (Ours)	60.00 $\pm$ 0.82	9.70 $\pm$ 6.42	55.00 $\pm$ 0.84	10.76 $\pm$ 8.33	43.33 $\pm$ 0.83	12.98 $\pm$ 9.97	41.67 $\pm$ 0.83	13.74 $\pm$ 9.58	41.67 $\pm$ 0.83	15.83 $\pm$ 12.34
SAVR-RGBImage+Force	26.67 $\pm$ 0.74	17.83 $\pm$ 10.23	28.33 $\pm$ 0.76	17.03 $\pm$ 10.55	25.00 $\pm$ 0.73	21.72 $\pm$ 15.06	21.67 $\pm$ 0.69	29.03 $\pm$ 20.01	21.67 $\pm$ 0.69	27.91 $\pm$ 19.55
SAVR-ForceOnly	60.00 $\pm$ 0.82	9.24 $\pm$ 6.73	50.00 $\pm$ 0.84	11.93 $\pm$ 9.87	40.00 $\pm$ 0.82	15.67 $\pm$ 12.28	41.67 $\pm$ 0.83	14.33 $\pm$ 10.40	40.00 $\pm$ 0.82	15.90 $\pm$ 11.90
SAVR-SegImageOnly	48.33 $\pm$ 0.84	13.67 $\pm$ 8.47	33.33 $\pm$ 0.79	13.11 $\pm$ 7.92	33.33 $\pm$ 0.79	16.06 $\pm$ 10.94	35.00 $\pm$ 0.80	13.59 $\pm$ 8.35	50.00 $\pm$ 0.84	13.17 $\pm$ 10.01

Note: Bold indicates the highest success rate and lowest average error for each target amount. Each approach was trained using three randomly selected seeds. For each trained model, evaluation was conducted over 20 trials per target amount using the same test seed, and results were averaged across the three trained models. SR denotes success rates, defined as scooping within 10% of the target amount; SD denotes standard deviation.

V. REAL-WORLD EXPERIMENTS

To evaluate the sim-to-real transfer ability of our approach, we deployed SAVR on a real robotic arm and conducted experiments on food scooping on three different types of food.

A. Experiment Setup and Implementation Details

The xArm-6 robotic arm (UFactory, China) was used for execution, with an IKEA Fornuft stainless steel spoon securely mounted onto a 6-axis force-torque sensor (M3712B, Sunrise Instruments, China). A Realsense D435i RGB-D camera (Intel, USA) was mounted on the gripper of the robotic arm, providing visual feedback during scooping (Fig. 1b). To estimate the food pose in 3D, we used GroundingDINO [27] to detect the bowl center in 3D, then deprojected the 2D point into 3D using depth information. The food pose was used to compute the initial position y_0 of the DMP, allowing the system to adjust for variations in bowl placement and food levels. To obtain the segmented spoon and food images in the real world, we used GroundingDINO to detect the bounding box of the spoon and food, allowing us to isolate the relevant region for segmentation. SAM2 [28] was then applied to track and mask the spoon and food through the scooping process.

B. Experiment Protocol

We conducted experiments on three different food types, each chosen to represent a range of textures encountered in robot-assisted feeding. Uncooked green beans represented small, hard food particles; corn kernels represented slightly larger, softer food particles that may deform under force; and mashed potatoes represented deformable, semi-solid foods.

To evaluate SAVR’s ability to generalize to different portion sizes, we tested its performance across four distinct target amounts: 25%, 50%, 75%, and 100% of the maximum scooped amount (Fig. 3). While the model was originally trained on 10% increments, we opted for 25% increments in real-world experiments. This was because in an assistive feeding scenario, users are unlikely to perceive differences as fine as 10% (Fig.



Fig. 2. Differences in scooped amounts with $\sim 10\%$ increments are visually subtle.

2), and a four-level granularity provides sufficient variation in portion sizes while keeping the evaluation process efficient.

To quantify the amount of food scooped, we used a precision electronic weighing scale (EMB 1200-1, KERN, Germany). The weight of the bowl was measured before and after each scooping action, with the difference providing a measurement of the actual scooped amount. To determine the maximum scooped amount, the experimenter performed five manual scooping trials, attempting to scoop as much as possible each time. The average weight across these trials was used to establish the 100% reference value for portion size calculations. To maintain consistency across trials, the scooped food was returned to the bowl, and the food surface was flattened after each scoop. This step minimized variations in food distribution that could otherwise introduce bias into the results.

We compared SAVR against DMP, SAVR-SegImageOnly and SAVR-ForceOnly for the real-world experiments. PPO-Only and SAVR-RGBImage+Force were excluded from real-world testing due to poor performance in simulation. For each approach, we selected the best-performing model from the simulation experiments for real-world testing. Each approach was tested across three trials per target amount, and the results were averaged over these trials. The same evaluation metrics detailed in Section IV-C were used.

TABLE IV
REAL-WORLD EXPERIMENT RESULTS (10% THRESHOLD)

Approach	25% Target		50% Target		75% Target		100% Target		Overall	
	SR $\pm$ SD (%) $\uparrow$	Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Error $\pm$ SD (%) $\downarrow$	SR $\pm$ SD (%) $\uparrow$	Error $\pm$ SD (%) $\downarrow$
DMP	44.44 $\pm$ 0.53	16.57 $\pm$ 15.99	22.22 $\pm$ 0.44	15.72 $\pm$ 9.42	0.00 $\pm$ 0.00	42.93 $\pm$ 8.38	0.00 $\pm$ 0.00	68.58 $\pm$ 10.31	22.22 $\pm$ 0.38	35.95 $\pm$ 24.64
SAVR (Ours)	44.44 $\pm$ 0.53	10.94 $\pm$ 7.65	22.22 $\pm$ 0.44	13.47 $\pm$ 8.55	33.33 $\pm$ 0.50	14.54 $\pm$ 9.17	44.44 $\pm$ 0.53	12.20 $\pm$ 6.99	36.11 $\pm$ 0.49	12.79 $\pm$ 7.90
SAVR-SegImageOnly	0.00 $\pm$ 0.00	23.34 $\pm$ 2.65	0.00 $\pm$ 0.00	46.89 $\pm$ 3.49	0.00 $\pm$ 0.00	51.99 $\pm$ 15.70	11.11 $\pm$ 0.33	45.05 $\pm$ 22.80	2.78 $\pm$ 0.17	41.82 $\pm$ 17.41
SAVR-ForceOnly	33.33 $\pm$ 0.50	26.58 $\pm$ 23.37	11.11 $\pm$ 0.33	22.78 $\pm$ 10.22	11.11 $\pm$ 0.33	30.62 $\pm$ 15.94	33.33 $\pm$ 0.50	35.12 $\pm$ 26.20	22.22 $\pm$ 0.42	28.77 $\pm$ 19.63

Note: Bold indicates the highest success rate and lowest average error. Each approach was tested over three trials per target amount using the best-performing model from simulation. SR denotes success rates, defined as scooping within 10% of the target amount; SD denotes standard deviation

TABLE V
REAL-WORLD EXPERIMENT RESULTS (20% THRESHOLD)

Approach	25% Target	50% Target	75% Target	100% Target	Overall
	SR $\pm$ SD (%) $\uparrow$	SR $\pm$ SD (%) $\uparrow$	SR $\pm$ SD (%) $\uparrow$	SR $\pm$ SD (%) $\uparrow$	SR $\pm$ SD (%) $\uparrow$
DMP	66.67 $\pm$ 0.50	55.56 $\pm$ 0.53	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	30.56 $\pm$ 0.47
SAVR (Ours)	88.89 $\pm$ 0.33	77.78 $\pm$ 0.44	77.78 $\pm$ 0.44	88.89 $\pm$ 0.33	83.33 $\pm$ 0.38
SAVR-SegImageOnly	11.11 $\pm$ 0.33	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	22.22 $\pm$ 0.44	8.33 $\pm$ 0.28
SAVR-ForceOnly	44.44 $\pm$ 0.53	33.33 $\pm$ 0.50	33.33 $\pm$ 0.50	33.33 $\pm$ 0.50	36.11 $\pm$ 0.49

Note: Bold indicates the highest success rate, defined as scooping within 20% of the target amount

C. Results and Discussion

Table IV presents the success rates and the amount errors in percentage for the real-world experiments. SAVR consistently outperforms the baselines in both success rate and amount error, demonstrating its ability to adapt to different food properties and scoop a target amount effectively.

SAVR achieves the highest overall success rate of 36.11% when using a 10% success threshold. While the success rate may seem moderate, the average amount error is only 12.79% $\pm$ 7.90%, indicating that the robot frequently scoops close to the target amount, with most failures occurring due to minor deviations from the strict 10% criterion. In practical applications, such small deviations are unlikely to be perceptible in a real feeding scenario. If the success criterion is relaxed to a 20% threshold, as shown in Table V, SAVR’s success rates increase to 83.33%, which is significantly more than the other baselines. This suggests that the model is still highly effective at producing scooped portions that are acceptably close to the target. In contrast, DMP and other ablated versions do not exhibit the same degree of improvement.

DMP achieves comparable success rates to SAVR for lower target amounts of 25% and 50%, but fails completely at higher amounts, suggesting that it lacks the adaptability required to handle different food textures and precise scooping due to its open-loop nature. The ablated versions of SAVR exhibit significantly lower success rates and higher errors, highlighting the necessity of both force and segmented image feedback for sim-to-real transfer. Notably, SAVR-ForceOnly outperforms SAVR-SegImageOnly, suggesting that force sensing plays a more critical role in accurate scooping, especially in real-world settings where image segmentation may be affected by lighting variations and domain gaps between real food textures and simulated particle-based representations.

Overall, our sim-to-real transfer results exceeded expectations, demonstrating that the learned policy generalized effectively

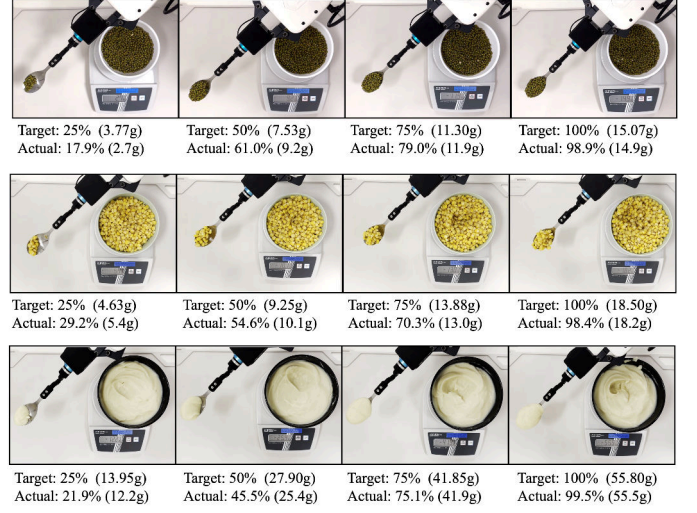


Fig. 3. Scooped amounts for each target level across different food types: (top) green beans, (middle) corn, and (bottom) mashed potatoes. Due to its viscous and adhesive texture, mashed potatoes heap significantly more at 100% target weight compared to the more granular green beans and corn, resulting in larger volumetric portions.

tively to real-world conditions despite the inherent differences between simulation and reality. We attribute this strong transfer performance to several factors: MuJoCo’s realistic force and contact modeling, and the use of segmented image feedback. Notably, the segmented spoon image provided a crucial cue for estimating the amount of food scooped, significantly reducing the sim-to-real gap since it relied on area rather than complex texture or color-based features.

VI. CONCLUSION AND FUTURE WORK

This work presents SAVR, a reinforcement learning-based framework for adaptive food scooping, enabling robots to dynamically adjust scooped amounts based on food properties and user-specified bite sizes. By integrating force feedback and segmented image masks, SAVR achieves high adaptability and accuracy in both simulation and real-world experiments, outperforming baselines in scooping precision and success rates. While the strict 10% threshold leads to moderate success rates, the consistently low absolute error indicates that SAVR reliably scoops close to the target amount.

Despite these promising results, several challenges remain. One key limitation is that our evaluation method defines the “maximum scoopable amount” based on the heaped capacity

of the spoon, which introduces inconsistencies across different food types. For example, while SAVR successfully transferred the scooping strategies from simulation to real foods to scoop a target amount by weight, this often resulted in excessive volumetric portions for soft foods such as mashed potatoes, where heaping occurs (Fig. 3). As humans typically perceive bite sizes volumetrically rather than by weight, future work should explore volumetric measurement techniques, such as depth-based reconstruction or density-aware estimation, to standardize bite-size validation. Additionally, our simulation environment models only discrete particle-based foods, limiting its ability to learn appropriate scooping strategies for soft foods, deformable foods. Incorporating deformable food physics into simulation, alongside improved volumetric estimation, could help the model learn more appropriate strategies for soft foods, enhancing bite-size consistency and sim-to-real transfer.

Overall, SAVR demonstrates promising results for bite-size personalization in robot-assisted feeding. With its strong sim-to-real transfer, this framework lays the groundwork for more intelligent and user-adaptive feeding systems, bringing robot-assisted feeding systems closer to practical deployment.

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